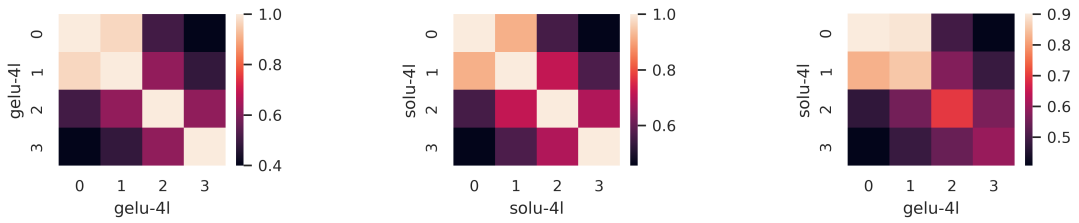

(a) (b) (c)

Figure 5: CKA for four layer GeLU and SoLU models (higher means more similar) evaluated on the Pile dataset (Gao et al., 2020). **(a, b)** Inter-model CKA for GeLU (respectively SoLU) models with 3 transformer blocks. **(c)** Between-model CKA for 3-block GeLU,SoLU models (rows correspond to SoLU layers, columns to GeLU layers.

(a) (b) (c)

Figure 6: Learning linear stitching layers for the stitching GeLU and SoLU models in Figure 1.

two distinct clusters in a natural metric space of hidden features of MNLI datapoints; for the technical definition of this metric space we refer to appendix C.

In fig. 3 we saw high CKA values between intermediate features of `pythia-{1b,1.4b}` and `pythia-6.9b`, but comparatively *low* CKA values between intermediate features of the 2.8 and 6.9 billion parameter models. Figure 9 shows that more generally there is a high level of intermediate feature similarity for models in the set `pythia-{1b,1.4b,6.9b}` but a low level of intermediate feature similarity *between* `pythia-2.8b` and any of the models in `pythia-{1b,1.4b,6.9b}`.

Table 1 suggests one hypothesis for why `pythia-2.8b` possesses unique hidden features from the perspective of CKA, namely that its attention head dimension is quite a bit smaller than those of the `pythia-{1b,1.4b,6.9b}` models.

model_name	d_head	rotary_dim
pythia-70m-deduped	64	16
pythia-160m-deduped	64	16
pythia-410m-deduped	64	16
pythia-1b-deduped	256	64
pythia-1.4b-deduped	128	32
pythia-2.8b-deduped	80	20
pythia-6.9b-deduped	128	32
pythia-12b-deduped	128	32

Table 1: Select HuggingFace (Wolf et al., 2020) model configuration dictionary keys for the Pythia models, showing that the 2.8b model has an unexpectedly small attention head dimension and hence rotary embedding demension (the latter appears to be tied to 1/4 of the former).

We make a couple final notes on our analysis of Pythia models as it relates to prior work on image models. First, restricting attention to intra-model CKA heatmaps in fig. 10 we see an increase in intra-

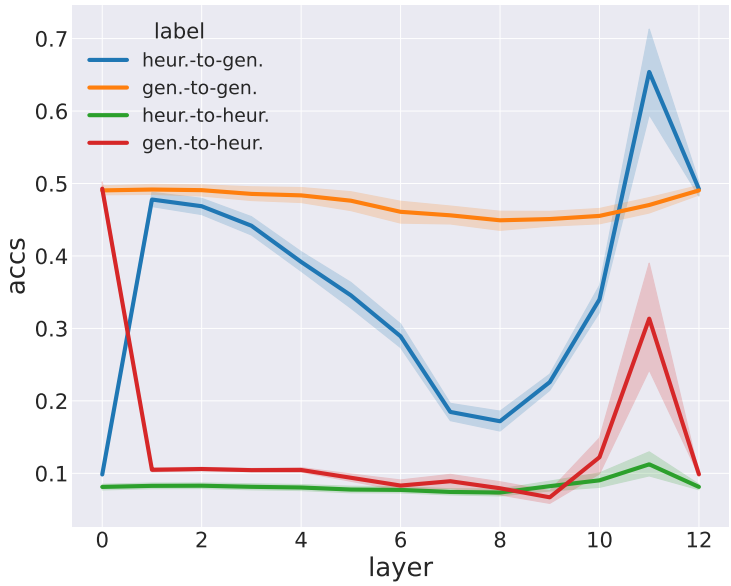

Figure 7: Identity stitching on the HANS-LO subset. The "heur.-to-heur." and "gen.-to-gen." models still exhibit distinct behavior.

model feature similarity as scale increases to 1 billion parameters,[4] analogous to an observation of (Nguyen et al., 2021). Note however that our experimental set up is not as clean as theirs: in our case both depth and width are scaled simultaneously, and perplexity of the Pythia models decreases as parameter count increases. In particular, our results would also be consistent with a hypothesis that intra-model feature similarity increases as validation loss decreases. Second, we speculate that the large blocks of intermediate features with high CKA similarity figs. 3 and 9 are related to the finding of (Raghu et al., 2022) that (compared to ResNets) Vision Transformers exhibit a high degree of feature uniformity across model layers. In other words, we hypothesize that this phenomenon is not specific to Vision Transformers, but rather transformer models more generally. However, as we only experiment with transformer models we are unable to make such a comparative statement and leave a broader study including e.g. RNNs and/or state-space sequence models to future work.

[4]See also relevant subplots of figs. 3 and 9 for larger model scales.

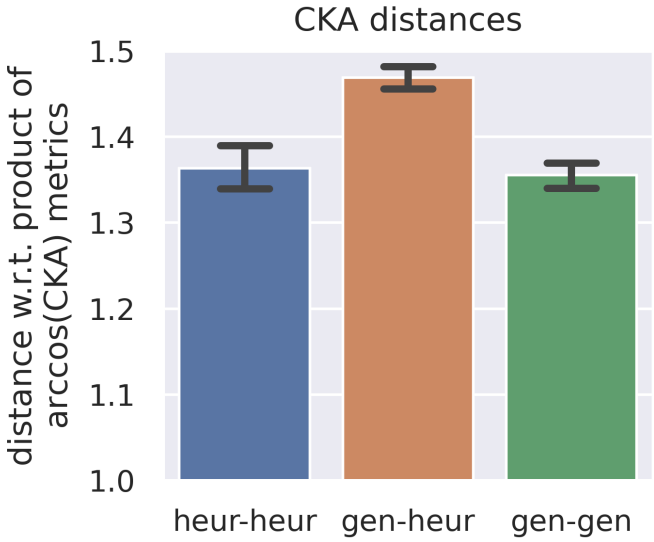

Figure 8: CKA-derived distances between the models of (McCoy et al., 2020) (higher means *less* similar, see appendix C for details on this distance metric). *Top-top*: pairwise CKAs between the top 10 models on HANS-LO. *Top-bottom*: pairwise CKAs between the top 10 and bottom 10 models on HANS-LO. *Bot-bot*: pairwise CKAs between the bottom 10 models on HANS-LO.